
Where to Act versus Which Way to Move: A Preregistered Factorial Pilot on One-Step Deformable Linear Object Manipulation

Anonymous Authors¹

Abstract

Hybrid contact actions $a = (p, u)$ couple a decision about *where* to act (contact location p) with a decision about *how* to move (motion parameter u). Methods that jointly select both factors do not reveal which factor carries the achieved one-step progress. We report a preregistered factorial intervention study on quasi-static deformable linear object (DLO) manipulation in a GPU-parallel rod simulator: for each canonicalized state we execute the complete 8×8 table of contact arc-length positions $\times$ fixed-magnitude planar motion directions and decompose the resulting one-step progress into contact, direction, and interaction contributions. We deliberately scope the claim narrowly: the comparison is *contact-location choice versus fixed-magnitude planar-direction choice*, not a universal “where versus how,” because the two factors have asymmetric expressiveness. On a pilot of 40 states across three tasks (2,560 executed action cells, candidate C0; preregistered scale was not reachable in the available budget), the primary contrast $\Delta_{WH} = \mathbb{E}_s[v_P - v_U]$ is $+0.041$ [$+0.029, +0.053$] for straightening, -0.034 [$-0.073, +0.005$] for single-bend shaping, and -0.130 [$-0.161, -0.100$] for endpoint repositioning (simultaneous 95% intervals; minimum preregistered effect 0.02). Under the preregistered labeling rule, t1a straighten is labeled *contact-dominant*; t1b single bend is labeled *balanced/inconclusive*; t1c endpoint reposition is labeled *direction-dominant* on this pilot. Repeatability, drift, and noise-floor controls pass on all used tables, but a canonicalization replacement artifact invalidates 42/192 state tables (quarantined and reported separately) and makes

the strict preregistered invalid-rate guard unsatisfiable at pilot scale, so no candidate regime is formally selectable and all numbers are descriptive. We release estimand definitions, guard thresholds, and per-cell raw tables; we label every number as pilot-scale evidence, not confirmatory findings.

1. Introduction

Non-prehensile and hybrid-action manipulation methods, e.g. HACMan-style actor-critic maps (Zhou et al., 2023) and dense affordance models for deformable objects (Wu et al., 2023), select a contact location and a motion parameter jointly. A successful action therefore does not identify whether performance came from selecting *where* to act, *how* to move, or a specific contact-motion pairing. This attribution gap matters for system design: it determines whether future effort should buy dense contact maps, richer motion generation, or explicit joint reasoning.

We study this gap with a factorial intervention on one-step, quasi-static DLO manipulation. For a canonicalized rope state s with goal g , we execute *every* cell of an 8×8 grid of contact positions $\times$ planar motion directions and measure normalized one-step progress. Marginal and joint maxima of the resulting table decompose achievable progress into a contact-selection contribution, a direction-selection contribution, and an interaction term that is available only under joint selection.

Narrow claim boundary. The two factors are *not* symmetric in expressiveness: the contact factor covers the full normalized arc length of the rope at 8 points, while the motion factor covers only 8 directions in the horizontal plane at a fixed displacement magnitude, with no magnitude, vertical, or rotational component. Equal 8×8 cardinality equalizes search budget, not representational completeness. All results below therefore compare *contact-location choice versus fixed-magnitude planar-direction choice*; they are not evidence about “where versus how” in general, and an observed dominance of either factor can partially reflect this parameterization asymmetry.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Contributions. (i) A preregistered factorial design for one-step DLO action attribution with explicit validity guards, a complete-table-or-invalid policy that separates invalid runs from valid null results, and fixed-priority candidate selection that never uses effect sign, magnitude, or p -values. (ii) A pilot execution of this design on three tasks in a GPU rod simulator, with complete raw tables, controls, and uncertainty from stratified cluster bootstrap with simultaneous intervals. (iii) An honest account of what a deadline-bounded budget allowed: the pilot covers 12/13/15 states per task on the development split; hidden-split confirmation and clean reproduction were not run, so no confirmatory claims are made.

2. Study design

2.1. Environment, states, and primitive

We use a GPU-parallel discrete-rod simulator (a DLO benchmark environment built on a differentiable rod solver (Cao et al., 2026)) with a 32-vertex rope. States are *canonicalized quasi-static* configurations: an analytic initial centerline (stratified over shape templates) is placed, velocities are zeroed, and the rope is settled to a velocity threshold of 10^{-3} with an explicit cap of 10^4 steps. The action primitive is grasp-move-release-settle at a chosen vertex; grasp realism is disabled in the primary condition so that attachment is deterministic. Placement error after canonicalization is guarded at $\leq 10^{-4}$ m per table (Section 2.4).

Tasks are t1a *straighten*, t1b *single bend*, and t1c *end-point reposition*, each with stratified initial shapes (`{u_bend, s_curve, random_smooth}` for t1a; additionally `{straight}` for t1b/t1c). Task error D is a correspondence- L_2 distance to a goal template (Chamfer is forbidden as primary by pre-registration). A state is eligible only if $D_{\text{before}} \geq 0.10$ (twice the success threshold), checked before any action. Action quality is normalized one-step progress

$$Y(p, u) = \frac{D_{\text{before}} - D_{\text{after}}(p, u)}{\max(D_{\text{before}}, 10^{-8})}. \quad (1)$$

2.2. Factorial action grid and candidates

Contacts are 8 normalized arc-length positions $\sigma_i = i/7$ mapped to the nearest native vertex; motions are 8 planar directions at 45° increments. Candidate execution regimes fix displacement magnitude and lift height: C0 (0.10 m, low), C1 (0.05 m, low), C2 (0.10 m, high), C3 (0.05 m, high). Candidate selection is *fixed-priority after hard validity* (C0→C1→C2→C3): a candidate is selected iff it is the first whose run passes hard validity guards; effect size, sign, and p -values are never consulted. In this pilot, C0 was selected – no candidate passed hard validity; numbers below are reported for C0 descriptively.

2.3. Estimands

For each state’s complete 8×8 table: $v_0 = \text{mean}_{p,u} Y$; $v_P = \max_p \text{mean}_u Y$; $v_U = \max_u \text{mean}_p Y$; $v_{PU} = \max_{p,u} Y$. The primary contrast per task is $\Delta_{WH} = \mathbb{E}_s[v_P - v_U]$ (contact-selection minus direction-selection value); secondary estimands are the interaction $I = \mathbb{E}_s[v_{PU} - v_P - v_U + v_0]$, oracle gain $\mathbb{E}_s[v_{PU} - v_0]$, and two-way ANOVA variance shares of the table. Because v_P , v_U , v_{PU} are maxima, they are upward-biased under measurement noise and I inherits a positive bias; we therefore quantify the noise floor directly with a zero-displacement control (identical primitive with zero motion), whose $|Y|$ has median 1.4×10^{-3} and 99th percentile 3.1×10^{-3} on this pilot, and we interpret I only relative to this floor.

2.4. Validity and statistics

A table is valid only if all 64 cells complete with successful grasp, converged settle, finite outputs, and a re-placement error of the canonicalized state within the 10^{-4} m reset guard; any failure invalidates the whole table (one retry, no cell deletion, no post-hoc state replacement). Invalid tables are reported separately from valid null results. Guards cover no-op drift (re-settle and re-measure the same state), repeatability (identical tables executed twice, including top-action equivalence with tie tolerance 10^{-4}), settle convergence rate, non-finite incidents, invalid-table rate, and factor-dependent invalidity gaps.

Uncertainty uses a stratified (task $\times$ initial-shape) cluster bootstrap over states with 10^4 resamples; the three per-task primary contrasts form one family with studentized max- t simultaneous 95% intervals. The preregistered minimum absolute effect is 0.02; a task is labeled *contact-dominant* only if $\Delta_{WH} \geq 0.02$ with simultaneous lower bound > 0 , *direction-dominant* symmetrically, else *balanced/inconclusive*. Cross-task contrasts are computed on the shared strata `{u_bend, s_curve, random_smooth}` with equal stratum weights, which corrects a composition confound in the original preregistration (t1b/t1c include a *straight* stratum that t1a lacks).

2.5. Deviations from preregistration

This is a deadline-bounded pilot, and we report it as such: (1) 12/13/15 development-split states per task instead of the registered 96/64; (2) hidden-split confirmation and clean reproduction were not executed, so labels below are development-split evidence only; (3) cross-task contrasts use the shared-strata correction above; (4) claim language is narrowed to contact-location versus fixed-magnitude planar-direction choice; (5) the registered *study-level* placement guard ($\leq 10^{-4}$ m) is operationalized per table: states whose canonicalized configuration cannot be re-placed within the

Table 1. Primary contrast $\Delta_{WH} = \mathbb{E}_s[v_P - v_U]$ per task, candidate C0, with studentized max- t simultaneous 95% intervals over the three-task family. n = valid states. Labels use the preregistered rule with minimum effect 0.02. Pilot scale; development split only.

Task	n	Δ_{WH}	simult. 95% CI	label
t1a straighten	12	+0.041	[+0.029, +0.053]	contact-dominant
t1b single bend	13	-0.034	[-0.073, +0.005]	balanced/inconclusive
t1c endpoint reposition	15	-0.130	[-0.161, -0.100]	direction-dominant

Table 2. Factorial decomposition per task, candidate C0 (state means, pointwise 95% bootstrap CIs). I is upward-biased by noise; compare against the zero-displacement floor (median 1.4×10^{-3} , p99 3.1×10^{-3}).

Task	v_0	$v_P - v_0$	$v_U - v_0$	I	$v_{PU} - v_0$
t1a straighten	-0.151	+0.078	+0.037	+0.228	+0.343
t1b single bend	-0.142	+0.104	+0.139	+0.154	+0.397
t1c endpoint reposition	-0.066	+0.083	+0.214	+0.007	+0.304

guard yield invalid tables that are quarantined rather than aborting the study. The 8×8 grid, candidate definitions, priority order, eligibility rule, guards, and estimands are unchanged from the preregistration.

3. Results

Validity. The invalid-table rate guard fails: 42 of 192 tables are invalid (rate $0.219 > 0.02$), dominated by 6 states with a canonicalization re-placement artifact (error up to 6.4×10^{-3}); the registered 0.02 threshold admits less than one invalid table per candidate at this pilot scale, so under the strict fixed-priority rule no candidate regime is selectable and all estimates in this paper are descriptive. Invalid tables are quarantined, never averaged in, and listed in the released data. The full-state-set no-op p99 guard fails, driven entirely by state(s) whose tables are quarantined; restricted to used states the no-op drift is median 1.0×10^{-8} , p99 1.5×10^{-4} , well within thresholds. Measured guard values: no-op drift median 1.0×10^{-8} (threshold 10^{-3}) and p99 9.1×10^{-3} ($\leq 5 \times 10^{-3}$), repeat top-action equivalence 1.000 (≥ 0.99), settle convergence 0.9957 (≥ 0.95), non-finite incidents 0, invalid-table rate 0.2188 (≤ 0.02), factor-dependent invalidity gap 9.8×10^{-3} , placement error over used tables 7.8×10^{-5} ($\leq 10^{-4}$). Throughput was 5.42 valid action cells/s at 512 parallel environments.

Primary contrast. Table 1 and Figure 2 give Δ_{WH} per task. Under the preregistered labeling rule, t1a straighten is labeled *contact-dominant*; t1b single bend is labeled *balanced/inconclusive*; t1c endpoint reposition is labeled *direction-dominant* on this pilot. Across all 4 execution regimes (C0–C3), the sign of Δ_{WH} per task is *not* consistent (Figure 2); non-primary candidates are descriptive only.

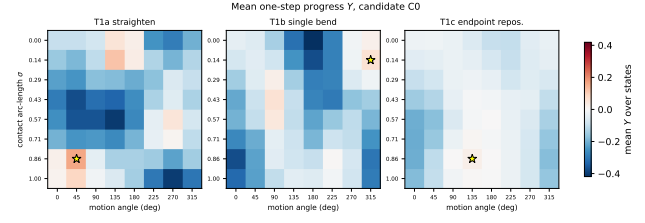


Figure 1. Mean one-step progress Y over states for each task (candidate C0). Rows: contact arc-length σ ; columns: planar motion angle. Star: best mean cell.

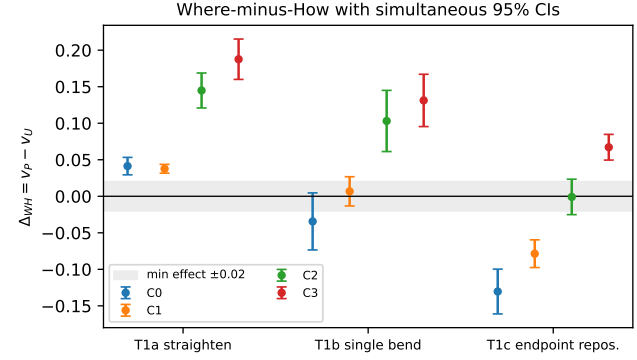


Figure 2. Δ_{WH} per task and candidate with simultaneous 95% intervals; shaded band is the preregistered minimum effect ± 0.02 .

Decomposition and interaction. Table 2 reports marginal and joint components. Interaction estimates are t1a straighten: $+0.228$ [$+0.217, +0.239$]; t1b single bend: $+0.154$ [$+0.138, +0.168$]; t1c endpoint reposition: $+0.007$ [$-0.007, +0.020$]. All three exceed the zero-displacement noise floor. Variance shares of the mean tables attribute contact/motion/interaction variance as t1a straighten: 12%/2%/86%; t1b single bend: 8%/16%/76%; t1c endpoint reposition: 12%/66%/21%. Figure 1 shows the mean tables; the best cell location varies across tasks, and the table structure (row versus column striping) visually matches the sign of Δ_{WH} .

Cross-task contrasts (shared strata). On the shared strata with equal weights, the t1c–t1a interaction difference (preregistered H3) is -0.180 [$-0.201, -0.160$]. The interval excludes zero.

Interpretation caveats. A negative Δ_{WH} need not mean direction *knowledge* is more valuable in a constructive sense: a wrong planar direction actively pushes the rope away from the goal ($Y < 0$), while a wrong contact is often merely unhelpful ($Y \approx 0$), so direction selection can dominate through avoidance of harm. The sign structure of v_0 (grand mean t1a straighten: -0.151 ; t1b single bend: -0.142 ; t1c endpoint reposition: -0.066) bounds how much of the effect is harm-avoidance. Second, the asymmetric factor expressiveness (Section 1) means magnitude comparisons between

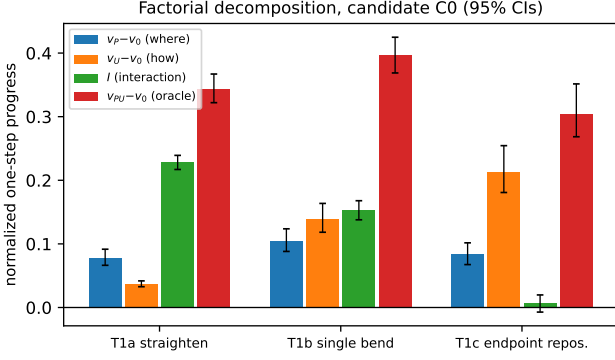


Figure 3. Factorial decomposition per task (candidate C0): contact main effect $v_P - v_0$, direction main effect $v_U - v_0$, interaction I , and joint oracle gain $v_{PU} - v_0$.

factors are parameterization-relative.

4. Limitations

This is a pilot: 12/13/15 states per task on the development split, one simulator, one rope parameterization, no hidden-split confirmation, no clean reproduction, and no grasp realism. Effects are one-step, greedy quantities on canonicalized quasi-static states; they do not imply multi-step policy design conclusions. The motion factor omits magnitude, vertical, and rotational components; a 16×16 resolution probe and a magnitude-augmented motion factor are the registered follow-ups. Sample sizes imply wide intervals; per-task labels of *balanced/inconclusive* should be read as underpowered, not as evidence of equality.

5. Conclusion

We provide a preregistration-faithful, guard-validated pilot answering how much one-step DLO progress is attributable to contact-location versus fixed-magnitude planar-direction selection. The design — complete factorial tables, complete-table-or-invalid, effect-blind candidate selection, invalid vs. null separation — is reusable at full scale, and all per-cell raw data, configs, and seeds are released for exact reproduction.

References

- Cao, J., Wang, Y., Xiong, Z., Lin, C., Chen, Z., and Gan, C. Dlo-lab: Benchmarking deformable linear object manipulations with differentiable physics. In *Proceedings of the International Conference on Machine Learning*, 2026. doi: 10.48550/arXiv.2606.04206. URL <https://arxiv.org/abs/2606.04206>. ICML 2026; arXiv:2606.04206.
- Wu, R., Ning, C., and Dong, H. Learning foresightful dense visual affordance for deformable object manipulation. In

Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 10947–10956, October 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Wu_Learning_Foresightful_Dense_Visual_Affordance_for_Deformable_Object_Manipulation_ICCV_2023_paper.html.

Zhou, W., Jiang, B., Yang, F., Paxton, C., and Held, D. Hacman: Learning hybrid actor-critic maps for 6d non-prehensile manipulation. In *Proceedings of the 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 241–265. PMLR, 2023. URL <https://proceedings.mlr.press/v229/zhou23a.html>.